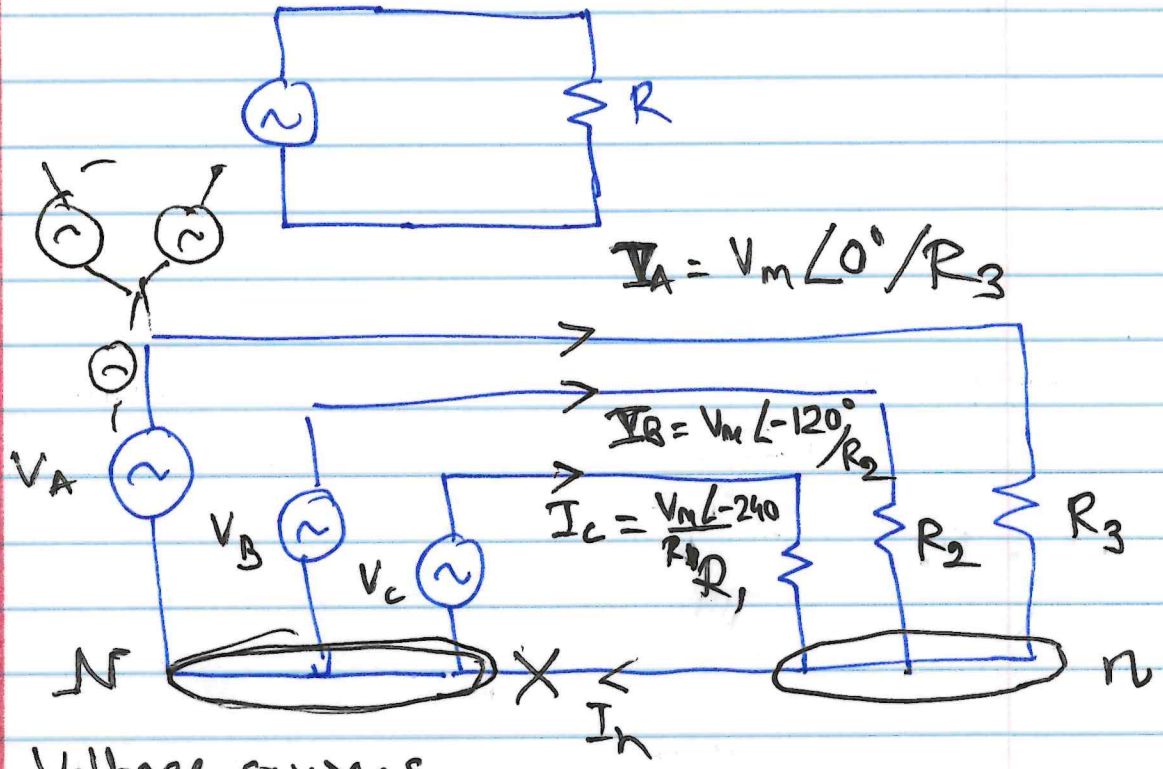
Josh Gage Meera.

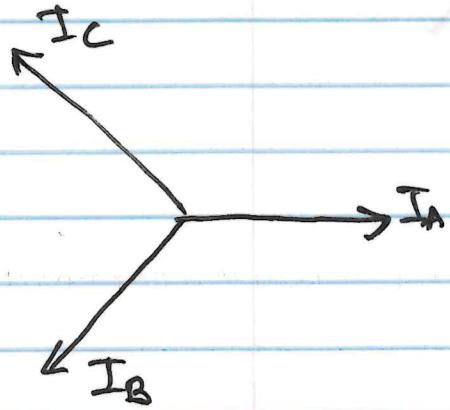
Lecture 5.

Singl phase → Three phase



Voltage sources

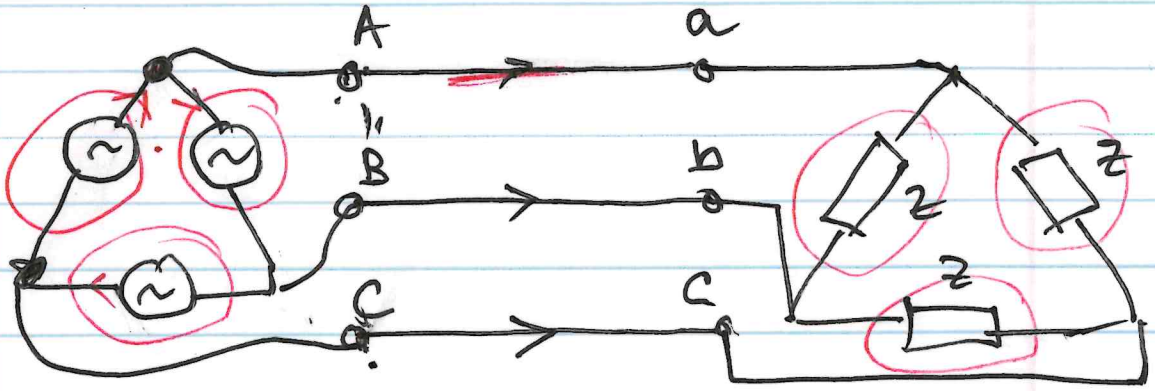
* 120° deg. apart.
* Magnitude
* Same frequency.

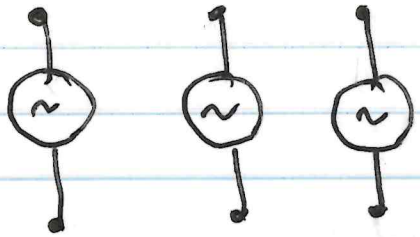


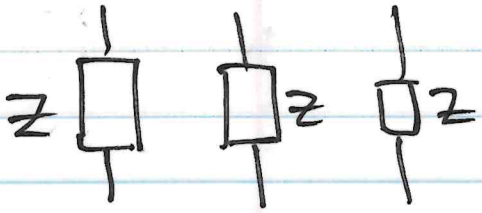
$$I_A = \frac{V_m}{R_3} \angle 0°$$

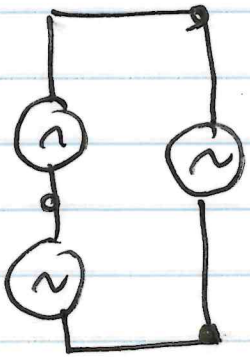
$$I_B = \frac{V_m}{R_2} \angle -120° \qquad R_1 = R_2 = R_3$$

$$I_C = \frac{V_m}{R_1} \angle -240°$$







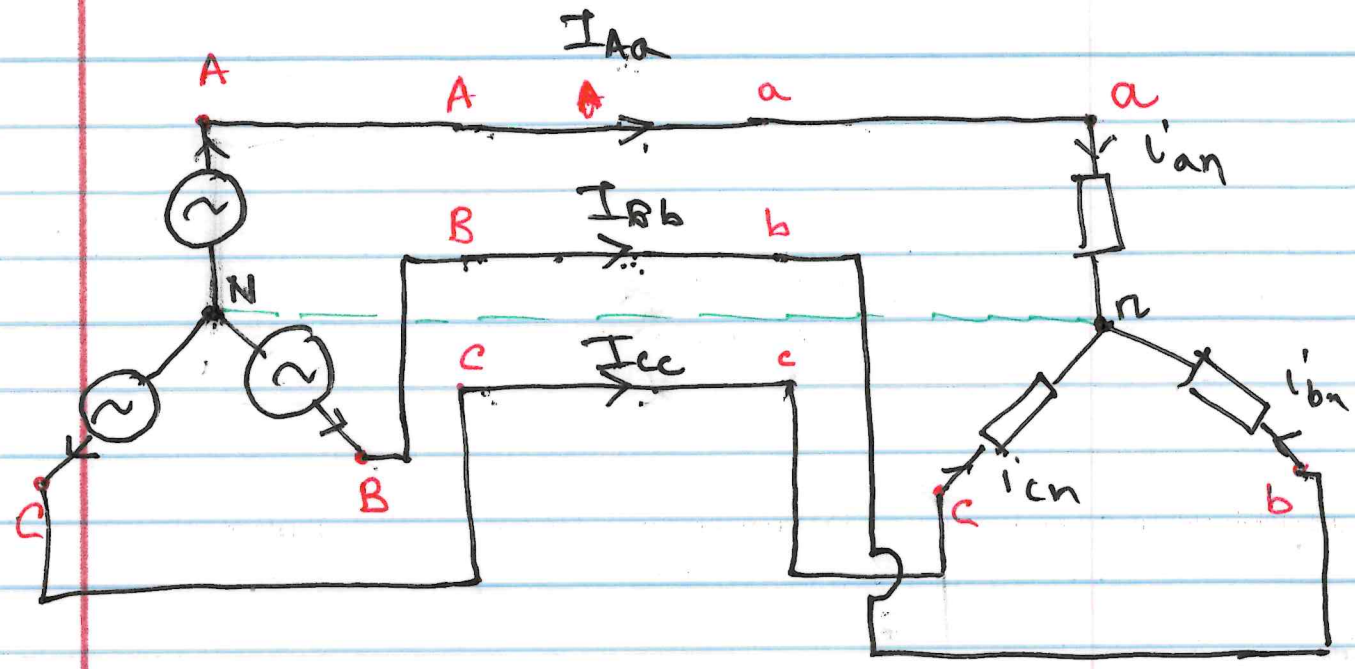


① Y – Y

③ Y – Δ

② Δ – Y

④ Δ – Δ

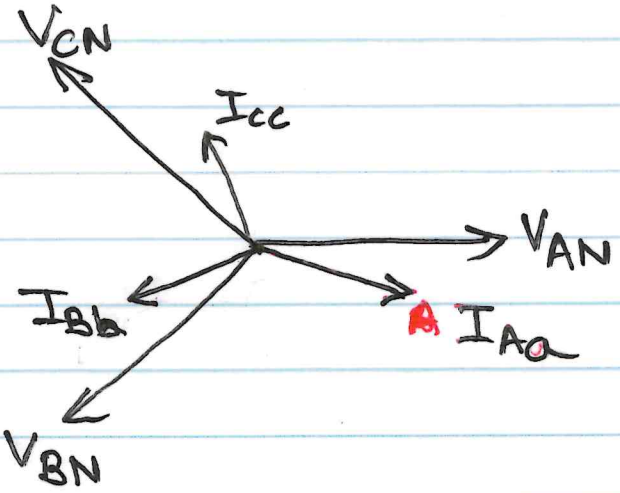


Line-to-line voltage → V_{AB}, V_{BC}, V_{CA}, V_{ab}, V_{bc}, V_{ca}

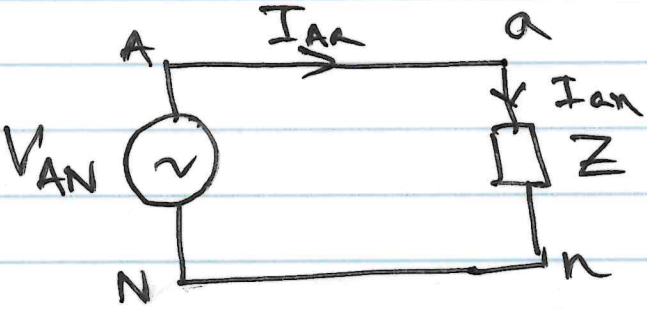
Phase voltages → V_{AN}, V_{BN}, V_{CN}, V_{an}, V_{bn}, V_{cn}.

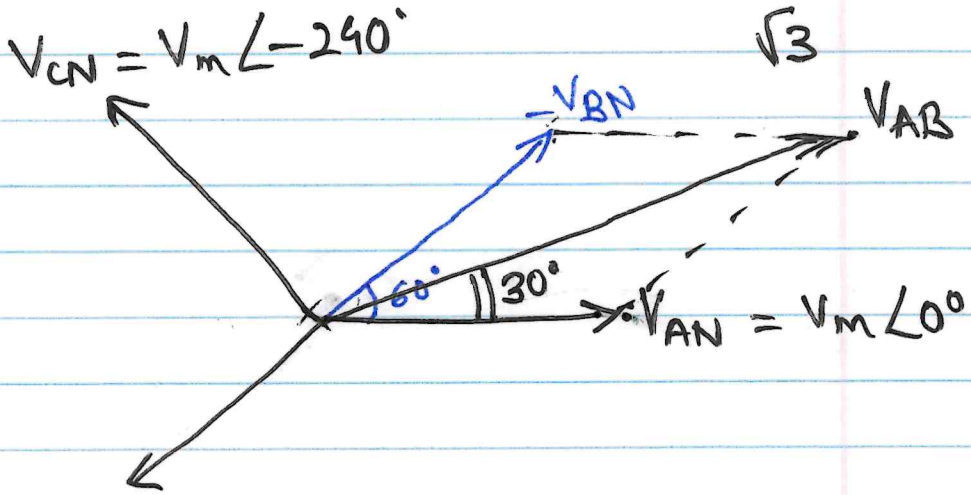
Phase Currents → $-I_{AN}$, $-I_{BN}$, $-I_{CN}$, i'_{an}, i'_{bn}, i'_{cn}.

Line Current → I_{Aa}, I_{Bb}, I_{cc}.



Single phase equivalent circuit.





$V_{BN} = V_m \angle -120°$

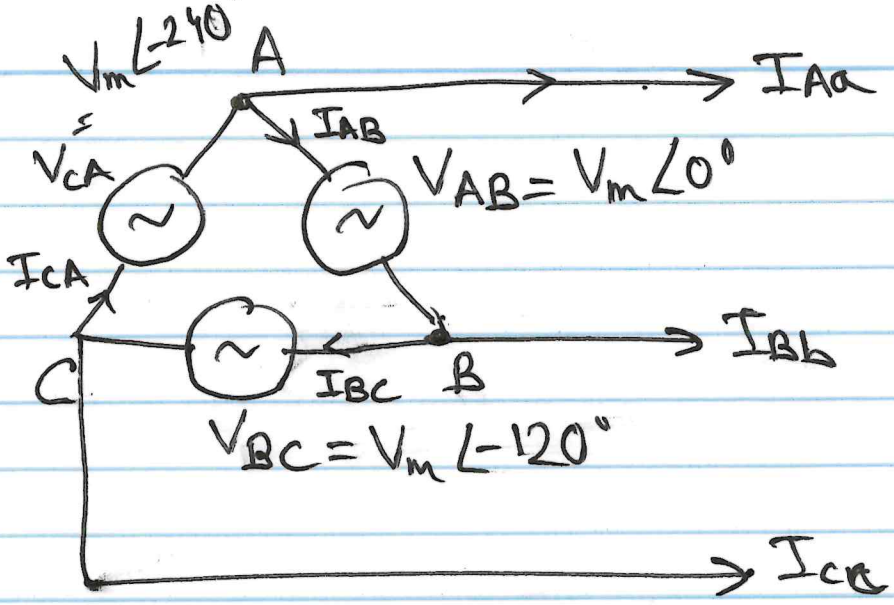
$V_{AB} = V_{AN} - V_{BN}$

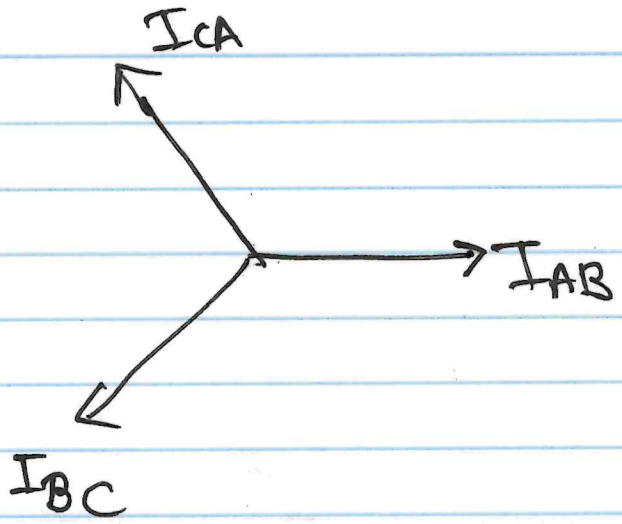
$= \sqrt{3}\, V_m \angle 30°$

$V_{AB} = ?$

$V_{Bc} = ?$

$V_{cA} = ?$


$V_m \angle -240°$
A
I_{AA}
I_{AB}
V_{CA}
$V_{AB} = V_m \angle 0°$
I_{CA}
I_{BB}
C
I_{BC}
B
$V_{BC} = V_m \angle -120°$
I_{CC}


I_{CA}
I_{AB}
I_{BC}